Conjecture 00000000821: no uniform gradient lower bound at the Laplace eigenvalue scale

lizaixi01, with OpenAI Codex assistance

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1 Source and the spectral convention

The source conjectures a positive uniform lower bound for

$$\frac{\max|\nabla\varphi_\lambda|}{\lambda\max|\varphi_\lambda|}.$$

It does not specify the underlying operator or its spectral convention. We use the standard positive Laplacian convention $-\Delta\varphi_\lambda = \lambda\varphi_\lambda$, with λ the eigenvalue itself. This is, for example, the convention used in the abstract of Steinerberger's paper [1]. Our conclusion concerns this interpretation. If λ instead means the square root of the eigenvalue, the statement is a different claim and the present argument does not refute it.

The counterexamples lie on one fixed compact manifold: the flat circle $M = \mathbb{R}/(2\pi\mathbb{Z})$ with arc-length metric $d\theta^2$. Thus even allowing the lower-bound constant to depend on this manifold does not rescue the assertion. In these coordinates the gradient norm is $|f'(\theta)|$ and the positive Laplacian is $-f''(\theta)$. These formulas follow directly from the constant metric coefficient $g_{11} = g^{11} = 1$. The usual sine/cosine spectrum of this circle is also recorded in Tsogtgerel's lecture notes [2], with the opposite sign convention for the Laplacian.

2 A family on the actual circle

For each integer $m \geq 1$ define

$$f_m([\theta]) = \sin(m\theta), \qquad D_m([\theta]) = m\cos(m\theta).$$

Both expressions are 2π-periodic, hence well-defined on M. They are continuous, their coordinate pullbacks are smooth, and

$$f_m' = D_m, \qquad D_m' = -m^2 f_m.$$

Consequently f_m is a nonzero smooth eigenfunction of the positive Laplacian, with eigenvalue $\lambda_m = m^2 > 0$.

The elementary bounds $|\sin t| \leq 1$ and $|\cos t| \leq 1$, together with the points $[\pi/(2m)]$ and $[0]$, give the attained maxima

$$\max_M |f_m| = 1, \qquad \max_M |\nabla f_m| = m.$$

No supremum is substituted for an unattained maximum. Therefore

$$\frac{\max_M |\nabla f_m|}{\lambda_m \max_M |f_m|} = \frac{m}{m^2} = \frac{1}{m}. \tag{1}$$

For every $c > 0$, the Archimedean property supplies $m \geq 1$ with $1/m < c$. Equation (1) rules out every positive constant uniform over eigenfunctions, including one uniform only on the high-frequency tail. Amplitude normalization has no effect on this ratio. The exhibited functions already have absolute maximum one, so the denominator-placement ambiguity in the Chinese text has no effect on this family.

3 What is formalized

The Lean object `FlatCircle` is Mathlib's actual additive quotient `AddCircle (2 * Real.pi)`. The functions `circleU` and `circleD` are defined by the periodic-function quotient lift. Their pullbacks are proved equal to the sine and cosine expressions; quotient-map continuity and smoothness of the first pullback are proved.

`CircleEigenpair` states positivity of the eigenvalue, continuity of both circle functions, a nonzero value, smoothness of the pullback, and two derivative identities in Lean's `HasDerivAt` predicate. It is the explicit arc-length coordinate formulation described above. We do not claim that the code invokes a general Riemannian gradient or Laplace–Beltrami API. The metric-to-coordinate formulas are spelled out here. No spectral, derivative, or maximum identity is assumed in the proof of the example.

Theorems `max_value` and `max_gradient` use `IsGreatest` on the value ranges of the actual quotient functions. The final theorem `no_uniform_positive_bound` negates the fully quantified positive bound for these nonzero circle eigenfunctions. Its proof constructs an arbitrary-small-ratio family, rather than using the zero function or a single finite-frequency discrepancy.

4 Verification and provenance

Lean 4.33.0 and Mathlib are pinned in the accompanying project. Full project build, source-bound axiom audit and official fresh-kernel replay are recorded in the evidence directory. Only the usual Lean foundational axioms are permitted; no admitted proof, native decision shortcut, or new axiom is used. The PDF was rendered and every page inspected.

The bilingual source is preserved from upstream commit:
`b355e42906e6eaf52bc7e9d005e0ef02d8aa7400`. OpenAI Codex assisted the mathematics, formalization and report. A separate supervising Codex agent reviewed the source and interpretation; this is AI review, not human certification. The family and its spectral behavior are standard; we make no claim of new mathematics. Organizer acceptance remains pending.

References

[1] S. Steinerberger, *On the Spectral Resolution of Products of Laplacian Eigenfunctions*, 2017. `https://arxiv.org/abs/1711.09826`.

[2] G. Tsogtgerel, *Math 319, Lecture 25*, McGill University, 8 March 2011, slide 2. `https://www.math.mcgill.ca/gantumur/math319w11/downloads/slides25.pdf`.